

RW4-0.4 Interface Control Document

Large Reaction Wheel - 0.4 Nms

Hardware Revision 3

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Prepared By:	AnyCompany Engineering Team
Approved By:	Systems Engineering

AnyCompany

1. Document History

Version	Date	Author	Description
1.0	2023-01-20	J. Morrison	Initial release for RW4-0.4 production baseline
1.3	2024-03-15	M. Chen	Updated thermal operating limits and derating curves
1.6	2025-10-05	Engineering Team	Added redundant RS-485 bus, 32-bit fault register, CRC32

1.1 Distribution List

Recipient	Organization	Copy Type
Systems Engineering	AnyCompany	Controlled
Flight Software	AnyCompany	Controlled
Quality Assurance	AnyCompany	Controlled
Customer Integration	External	Controlled

1.2 Reference Documents

ID	Document	Title
RD-1	AC-DS-00063	RW4-0.4 Product Data Sheet
RD-2	AC-UM-00063	RW4-0.4 User Manual
RD-3	EIA/TIA-422-B	Balanced Voltage Digital Interface Circuits
RD-4	EIA/TIA-485-A	Balanced Multipoint Systems
RD-5	NASA GSFC-STD-7000A	General Environmental Verification Standard
RD-6	ECSS-Q-ST-70-02C	Thermal Vacuum Outgassing Test
RD-7	MIL-DTL-83513	Micro-D Connector Specification

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3. Scope and Applicability

This Interface Control Document defines the mechanical, electrical, environmental, and software interfaces for the AnyCompany RW4-0.4 large-class reaction wheel, hardware revision 3. It establishes requirements and constraints for spacecraft integration.

The RW4-0.4 provides 0.4 Nms angular momentum storage with 100 mNm peak torque. It is designed for ADCS on small to medium spacecraft (50-500 kg). Communication is via RS-422 primary and RS-485 redundant interfaces on a 28V bus (22-34V). Nominal power consumption is 2.8W.

Flight heritage: 28 units across 2 generations with 50+ cumulative flight years since 2023. Electronics common with the RW4-1.0 model.

Applicable to HW revision 3, firmware v4.2.x or later.

4. Mechanical Interface

4.1 Mounting Interface

Eight (8) M4 threaded mounting holes on a 95 mm bolt circle. Alignment pin hole (3.0 mm dia, 4.0 mm depth) for angular clocking.

- Minimum standoff: 5.0 mm to prevent frame warping
- Fastener torque: 2.5 Nm (M4 socket head, A2-70 SS)
- Mounting surface flatness: 0.05 mm over bolt circle
- Alignment pin tolerance: H7/g6 fit

4.2 Mass and Dimensions

Parameter	Value	Units	Tolerance
Mass	755	g	+/- 15
Width (X)	105	mm	+/- 0.5
Depth (Y)	103	mm	+/- 0.5
Height (Z)	46.1	mm	+/- 0.3
Bolt Circle	95	mm	+/- 0.1
Mount Holes	8x M4	-	6mm depth
Alignment Pin	3.0 x 4.0	mm	H7

4.3 Outline Drawing

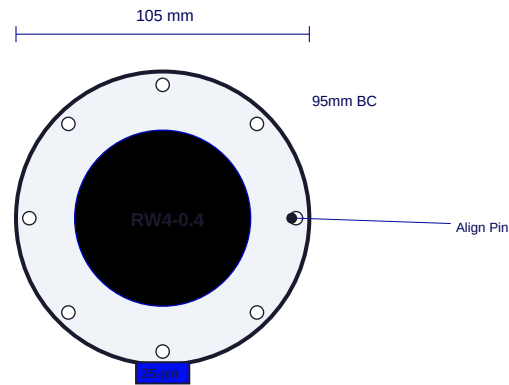


Figure 1: Top-view outline (mounting holes, alignment pin, connector).



Figure 2: RW4-0.4 product photograph.

4.4 Remove Before Flight

Item	Remove?	Notes
Kapton bearing covers	Yes	2x tape protecting bearings
Connector dust cap	Yes	Red anodized cap on 25-pin
Shipping restraint	Yes	Al bracket securing rotor

5. Environmental Specifications

5.1 Summary Table

Parameter	Min	Typ	Max	Units	Notes
Survival Temp	-45	-	+85	°C	Non-op, unpowered
Operating Temp	-30	+20	+70	°C	At baseplate
Storage Temp	-45	+20	+85	°C	Clean env, sealed bag w/ desiccant
Random Vib (Qual)	-	20.16	-	Grms	2 min/axis, 20-2000 Hz
Random Vib (Accept)	-	14.3	-	Grms	1 min/axis
Shock	-	-	500	G	Half-sine 0.5 ms, 3-axis
Radiation (TID)	-	-	50	krad	Board-level Si equiv
Pressure	0	-	101.3	kPa	Vacuum to sea level
TML	-	-	1.0	%	ECSS-Q-ST-70-02C
CVCM	-	-	0.1	%	ECSS-Q-ST-70-02C

5.2 Vibration Profile

Qualified to random vibration exceeding NASA GEVS (20.16 Grms).

Frequency	Qual Level	Accept Level	Duration
20-50 Hz	+6 dB/oct ramp	+3 dB/oct ramp	-
50-800 Hz	0.05 g ² /Hz flat	0.025 g ² /Hz flat	-
800-2000 Hz	-6 dB/oct rolloff	-6 dB/oct rolloff	-
Overall	20.16 Grms	14.3 Grms	2/1 min per axis

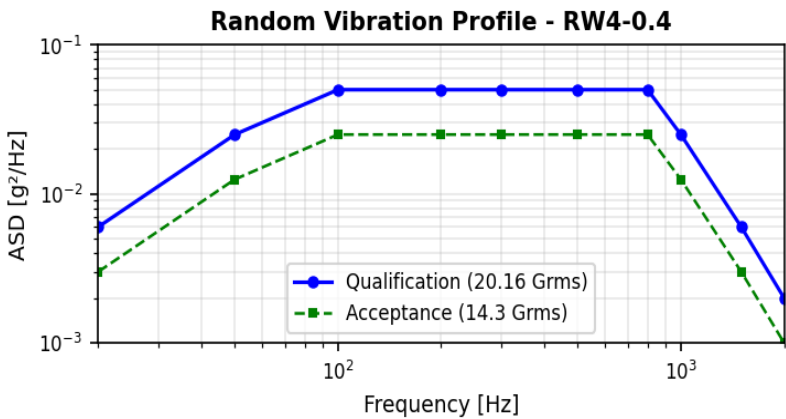


Figure 3: Random vibration PSD profile.

5.3 Shock and Radiation

Shock: 500G half-sine, 0.5 ms, three axes.

Radiation-hardened components: 50 krad (Si) TID. SEE mitigation via triple-modular redundancy and EDAC on memory.

Pressure: vacuum through sea level. Not qualified for high-altitude ascent. Outgassing: TML < 1%, CVCM < 0.1% per ECSS-Q-ST-70-02C.

6. Electrical Interface

6.1 Power Architecture

Dual-redundant power inputs with EMI filtering and DC/DC conversion. 28V bus (22-34V) regulated to internal 5V/3.3V. 3-phase BLDC motor driver with FOC. Regenerative braking with bus clamp at 36V.

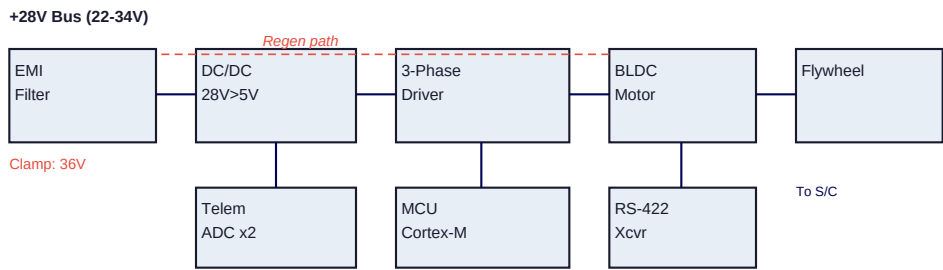


Figure 4: Power architecture block diagram.

6.2 Connector and Pinout

25-pin Micro-D (MIL-DTL-83513 compatible).

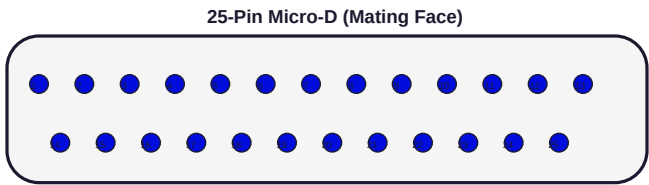


Figure 5: Connector face view (mating side).

Pin	Signal	Color	Description	Level
1	+28V_PRI	Red	Primary power input	22-34V
2	+28V_RED	Red/Wht	Redundant power input	22-34V
3	RTN_PRI	Black	Primary power return	GND
4	RTN_RED	Blk/Wht	Redundant power return	GND
5	CHASSIS	Green	Chassis/EMI ground	GND
6	TxA+	Orange	RS-422 Primary Transmit +	Diff
7	TxA-	Brown	RS-422 Primary Transmit -	Diff
8	RxA+	Blue	RS-422 Primary Receive +	Diff
9	RxA-	White	RS-422 Primary Receive -	Diff
10	TxB+	Violet	RS-485 Redundant Transmit +	Diff
11	TxB-	Gray	RS-485 Redundant Transmit -	Diff
12	RxB+	Pink	RS-485 Redundant Receive +	Diff
13	RxB-	Tan	RS-485 Redundant Receive -	Diff

Pin	Signal	Color	Description	Level
14	INHIBIT_A_N	Yellow	Active-low inhibit channel A	0/5V
15	INHIBIT_B_N	Yel/Wht	Active-low inhibit channel B	0/5V
16	TELEM_SYNC	Cyan	Telemetry sync pulse output	3.3V
17	ANALOG_OUT	Wht/Blu	Analog speed output 0-3.3V	Analog
18	HEATER_EN	Red/Grn	Heater enable control	0/5V
19-25	SPARE	-	Reserved for future use	-

6.3 Power Specifications

Parameter	Min	Nominal	Max	Units
Input Voltage	22	28	34	V
Idle Power (at speed)	-	2.8	3.5	W
Peak Power (100 mNm)	-	-	25	W
Inrush Current	-	-	5	A
Inrush Duration	-	-	20	ms
Bus Capacitance (total)	-	220	-	uF
Front-end Capacitance	-	1.0	-	uF
Regen Bus Clamp Voltage	-	36	-	V
ESD Tolerance (HBM)	-	-	+/-4	kV

6.4 Signal Specifications

Parameter	Min	Nominal	Max	Units
RS-422 Baud Rate	-	460800	-	bps
RS-422 Differential Output	2.0	-	5.0	V
RS-422 Input Sensitivity	0.2	-	-	V
RS-422 Line Termination	-	120	-	Ω
Maximum Cable Length	-	-	15	m
INHIBIT Input Low	-	-	0.8	V
INHIBIT Input High	2.0	-	5.0	V
ANALOG_OUT Range	0	-	3.3	V
TELEM_SYNC Pulse Width	-	10	-	μ s

7. Software Interface

7.1 Protocol Overview

Custom binary packet protocol over RS-422 (primary) or RS-485 (redundant). Supports command/reply transactions and periodic telemetry streaming. All multi-byte fields are little-endian. Sign convention: positive torque/speed = CCW viewed from the mounting face (connector side).

7.2 Packet Structure

Each packet begins with a 2-byte sync word (0xEB90), followed by length, message ID, sequence number, variable-length payload, and a 32-bit CRC.



Max frame: 1032 bytes (2+2+1+1+1024+4)

Figure 6: Binary packet frame structure.

CRC32 is computed over the entire frame excluding the CRC field itself, using polynomial 0x04C11DB7 with initial value 0xFFFFFFFF. The CRC is transmitted LSB first.

7.3 Message Identifiers

Code	Name	Direction	Description	Payload Size
0x01	HEARTBEAT	RW > SC	Periodic heartbeat (1 Hz)	4 bytes
0x02	SET_MODE	SC > RW	Set operating mode	1 byte
0x03	SET_SPEED	SC > RW	Speed setpoint (float32 RPM)	4 bytes
0x04	SET_TORQUE	SC > RW	Torque setpoint (float32 mNm)	4 bytes
0x05	SET_MOMENTUM	SC > RW	Momentum setpoint (float32 Nms)	4 bytes
0x10	TELEM_REQUEST	SC > RW	Request telemetry packet	1 byte
0x11	TELEM_HK	RW > SC	Housekeeping telemetry	64 bytes
0x12	TELEM_SCIENCE	RW > SC	Science telemetry	128 bytes
0x20	CONFIG_READ	SC > RW	Read config register	2 bytes (addr)
0x21	CONFIG_WRITE	SC > RW	Write config register	6 bytes (addr+val)
0x22	CONFIG_STORE	SC > RW	Store config to NVM	0 bytes
0x30	FW_UPDATE_START	SC > RW	Begin firmware update	8 bytes
0x31	FW_UPDATE_DATA	SC > RW	Firmware data block	up to 1024 bytes
0x32	FW_UPDATE_VERIFY	SC > RW	Verify firmware image	4 bytes (CRC)
0x40	SELF_TEST	SC > RW	Execute built-in self test	0 bytes
0x41	CALIBRATE	SC > RW	Run calibration routine	1 byte (type)
0xF0	ERROR	RW > SC	Error report	4 bytes (code)
0xFF	RESET	SC > RW	Software reset	0 bytes

7.4 Operating Modes

Mode ID	Name	Description
0x00	STANDBY	Motor unpowered, telemetry active, accepts all commands
0x01	SPEED_CTRL	Closed-loop speed control to RPM setpoint
0x02	TORQUE_CTRL	Open-loop torque command in mNm
0x03	MOMENTUM_CTRL	Closed-loop momentum control to Nms setpoint
0x04	SAFE	Reduced functionality, motor actively braked, fault recovery
0x05	FW_UPDATE	Firmware update mode (entered via FW_UPDATE_START)
0x06	SELF_TEST	Built-in self test (returns to STANDBY on completion)

Note: SAFE mode is entered automatically on critical fault. Invalid mode transitions return an ERROR packet. Minimum 2-second hold in STANDBY before entering active modes.

7.5 Command Details

SET_SPEED (0x03): 32-bit IEEE 754 float, desired wheel speed in RPM. Positive = CCW, negative = CW. Range: +/- 6500 RPM. Ramp rate configurable (default 500 RPM/s). Response: ACK or ERROR.

SET_TORQUE (0x04): float32 torque in mNm. Range: +/- 100 mNm. Applied immediately, no ramp. Duration limited by thermal protection.

SET_MOMENTUM (0x05): float32 momentum in Nms. Range: +/- 0.4 Nms. Internal PID controller adjusts motor torque to maintain target momentum.

SELF_TEST (0x40): Comprehensive BIST: (1) memory EDAC check, (2) ADC calibration, (3) motor driver continuity, (4) Hall sensor validation, (5) comm loopback. ~10 seconds. Results in TELEM_HK with fault register updated.

CONFIG_WRITE (0x21): Writes 32-bit value to register address. Changes are volatile until CONFIG_STORE (0x22). See Section 10 for register map.

TELEM_REQUEST (0x10): Requests a single telemetry packet. Payload byte: 0x01 = HK, 0x02 = Science. For periodic streaming, configure telem rates via CONFIG_WRITE to registers 0x0200/0x0204.

8. Telemetry Definitions

8.1 Housekeeping Telemetry (MSG_ID 0x11, 10 Hz)

The HK packet provides a comprehensive snapshot of wheel status at 10 Hz (configurable 1-100 Hz). Total packet size: 64 bytes.

Offset	Field	Type	Units	Range
0	wheel_speed	float32	RPM	+/- 6500
4	torque_actual	float32	mNm	+/- 120
8	current_phaseA	float32	A	0 - 5.0
12	current_phaseB	float32	A	0 - 5.0
16	current_phaseC	float32	A	0 - 5.0
20	bus_voltage	float32	V	0 - 40
24	bus_current	float32	A	0 - 2.0
28	pcb_temperature	float32	°C	-50 to +100
32	motor_temperature	float32	°C	-50 to +120
36	bearing_temperature	float32	°C	-50 to +100
40	fault_register	uint32	bitfield	See Section 9
44	mode_state	uint8	enum	0x00 - 0x06
45	reserved	uint8[3]	-	-
48	uptime	uint32	seconds	0 to 2 ³²
52	cumulative_revs	uint64	revolutions	0 to 2 ⁶⁴
60	power_consumed	uint32	mWh	0 to 2 ³²

8.2 Science Telemetry (MSG_ID 0x12, 100 Hz)

High-rate data for jitter characterization and control loop analysis. Total packet size: 128 bytes. Configurable 1-1000 Hz.

Offset	Field	Type	Units	Range
0	timestamp	uint64	microseconds	Monotonic clock
8	speed_highres	float64	RPM	High-resolution
16	torque_highres	float64	mNm	High-resolution
24	vibration_x	float32	mg	+/- 5000
28	vibration_y	float32	mg	+/- 5000
32	vibration_z	float32	mg	+/- 5000
36	current_total	float32	A	0 - 5.0
40	encoder_position	uint32	counts	0 - 65535
44-127	reserved	uint8[84]	-	-

9. Fault Management

The 32-bit fault register provides real-time fault status. Bits 0-15 are individual fault flags; bits 16-31 form a 16-bit event counter incremented on each new fault detection.

9.1 Fault Register Bitfield

Bit	Name	Description	Recovery Action
0	OC_PHASE_A	Overcurrent phase A (threshold: 5A)	Auto-retry x3, then SAFE
1	OC_PHASE_B	Overcurrent phase B (threshold: 5A)	Auto-retry x3, then SAFE
2	OC_PHASE_C	Overcurrent phase C (threshold: 5A)	Auto-retry x3, then SAFE
3	OV_BUS	Bus overvoltage (threshold: 36V)	Disable regen, enter SAFE
4	UV_BUS	Bus undervoltage (threshold: 20V)	Enter SAFE mode
5	OT_PCB	PCB overtemperature (threshold: 85°C)	Reduce torque limit to 50%
6	OT_MOTOR	Motor overtemp (threshold: 110°C)	Enter SAFE mode
7	OT_BEARING	Bearing overtemp (threshold: 90°C)	Enter SAFE mode
8	HALL_FAULT	Hall sensor error detected	Enter SAFE mode
9	ENCODER_FAULT	Incremental encoder error	Fallback to Hall commutation
10	COMM_TIMEOUT_PRI	Primary RS-422 timeout (>5s)	Switch to redundant RS-485
11	COMM_TIMEOUT_RED	Redundant RS-485 timeout	Enter SAFE mode
12	MOTOR_STALL	Motor stall (<10 RPM under torque)	Auto-retry, then SAFE
13	SPEED_RUNAWAY	Speed exceeds +/- 7000 RPM	Emergency brake, SAFE
14	MEMORY_FAULT	EDAC uncorrectable error	Enter SAFE, report via ERROR
15	WATCHDOG_RESET	Watchdog timer expired	Software reset
16-31	EVENT_COUNTER	16-bit fault event counter	Monotonic, non-resettable

9.2 Recovery Actions

On SAFE mode entry, the motor is actively braked to zero speed. All telemetry continues at the configured rate. The wheel remains in SAFE until a SET_MODE command transitions it back to STANDBY. A minimum 2-second hold in STANDBY is required before re-entering any active control mode.

Communication failover: if the primary RS-422 link is silent for 5 seconds, the wheel automatically activates the redundant RS-485 interface and sets the COMM_TIMEOUT_PRI fault bit. If both links fail, the wheel enters SAFE mode and remains there until communication is restored.

Fault register is read-only. Individual fault bits are cleared automatically when the fault condition is no longer present (except WATCHDOG_RESET which clears on next successful boot). The EVENT_COUNTER never resets and wraps at 65535.

10. Register and Memory Map

Configuration and status registers accessible via CONFIG_READ (0x20) and CONFIG_WRITE (0x21). All registers are 32-bit aligned. Write access requires the wheel to be in STANDBY mode unless otherwise noted.

Address	Field	Type	Access	Description
0x0000	speed_setpoint	float32	R/W	Speed command (RPM)
0x0004	torque_setpoint	float32	R/W	Torque command (mNm)
0x0008	momentum_setpoint	float32	R/W	Momentum command (Nms)
0x000C	mode_register	uint32	R/W	Operating mode (Section 7.4)
0x0010	speed_measured	float32	R	Current wheel speed (RPM)
0x0014	torque_measured	float32	R	Current torque output (mNm)
0x0018	bus_voltage	float32	R	Measured bus voltage (V)
0x001C	bus_current	float32	R	Measured bus current (A)
0x0020	pcb_temperature	float32	R	PCB temperature (°C)
0x0024	motor_temperature	float32	R	Motor temperature (°C)
0x0028	bearing_temperature	float32	R	Bearing temperature (°C)
0x002C	fault_register	uint32	R	Fault status (Section 9)
0x0030	uptime_seconds	uint32	R	Time since power-on (s)
0x0034	cumul_revs_lo	uint32	R	Cumulative revs (low 32 bits)
0x0038	cumul_revs_hi	uint32	R	Cumulative revs (high 32 bits)
0x003C	firmware_version	uint32	R	FW version (major.minor.patch)
0x0040	serial_number	uint32	R	Factory serial number
0x0044	hw_revision	uint32	R	Hardware revision code
0x0100	speed_kp	float32	R/W	Speed loop proportional gain
0x0104	speed_ki	float32	R/W	Speed loop integral gain
0x0108	speed_kd	float32	R/W	Speed loop derivative gain
0x010C	torque_limit	float32	R/W	Maximum torque (mNm)
0x0110	speed_limit	float32	R/W	Maximum speed (RPM)
0x0114	accel_limit	float32	R/W	Acceleration limit (RPM/s)
0x0200	telem_hk_rate	uint32	R/W	HK telemetry rate (Hz, 1-100)
0x0204	telem_sci_rate	uint32	R/W	Science telem rate (Hz, 1-1000)
0x0208	watchdog_timeout	uint32	R/W	Watchdog timeout (ms, 0=disable)
0x020C	comm_timeout	uint32	R/W	Comm failover timeout (ms)

11. Notes and Constraints

- The spacecraft power system must accommodate regenerative braking current. During deceleration, the wheel pushes current back onto the power bus. If the power line is disconnected (e.g., via relay), regeneration will increase the bus voltage until the 36V clamp activates and the wheel resets.
- Both INHIBIT lines must be deasserted (logic high) for the motor to operate. Asserting either INHIBIT immediately disables the motor driver and places the wheel in SAFE mode.
- The watchdog timer must be serviced by the spacecraft within the configured timeout (default 5 s). Failure to send any valid command within the timeout triggers a software reset.
- Maximum cable length for RS-422 is 15 meters. For cable runs exceeding 10 meters, shielded twisted-pair cable with drain wire connected to the CHASSIS pin is required.
- During firmware update mode, all motor control is disabled. The update process takes approximately 45 seconds for a complete firmware image. Power must not be interrupted during the update.
- The wheel must be stored in a clean environment to prevent bearing contamination. Use sealed ESD-safe bags with desiccant for long-term storage. Relative humidity should not exceed 50%.
- The wheel must not be powered during atmospheric ascent unless arcing risk has been verified by additional qualification testing for the specific altitude pressure profile.
- Mounting fasteners should be torqued in a star pattern to ensure even clamping force. Do not exceed the specified torque value of 2.5 Nm to avoid bearing preload distortion.

End of Document